

Smart Aeroponics System
An Engineering Project in Community Service

Phase – II Report

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***in partial fulfillment of the requirements for the degree of Bachelor of
Technology***



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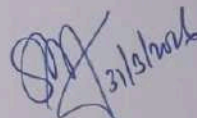


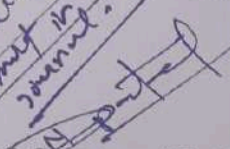
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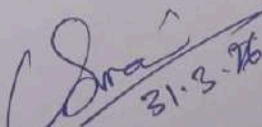
Bonafide Certificate

Certified that this project report titled **"Smart Aeroponics System"** is the bonafide work of **"23BAI10917 - Vijay Rajesh R, 23BAC10009 - Shivaraman T , 23BSA10102 - Ashwin Rajan R , 23BAS10021 - Agathiyan V , 23MEI10036 - M Thanushhri, 23BCY10293 - R Manoj Kumar "** who carried out the project work under my supervision.

This project report (Phase II) is submitted for the Project Viva-Voce examination held on 31st March 2026.


31/3/2026
Supervisor

Recommended to submit in journal.

Comments & Signature (Reviewer 1)


31.3.26.
Comments & Signature (Reviewer 2)



Declaration of Originality

We, hereby declare that this report entitled **Smart Aeroponics System** represents our original work carried out for the EPICS project as a student of VIT Bhopal University and, to the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section "References".

Date: 30/03/2026

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ABSTRACT

This Phase-I report details the design, implementation, and preliminary validation of a novel Smart Aeroponics System aimed at enabling cost-effective, sustainable, and highly intelligent soilless agriculture, particularly suited for urban and resource-constrained environments. The core system employs aeroponics with a high-frequency ultrasonic nebulizer to generate an oxygen-rich nutrient mist for optimal root delivery. Its intelligence is built upon continuous, real-time sensing of critical parameters including pH, TDS/EC, and temperature, managed by an ESP32 microcontroller which executes a robust closed-loop control strategy. This automation includes precise, iterative dosing of nutrients and pH solutions via peristaltic pumps, regulated misting cycle control, and a critical automatic low-water shutdown safety protocol. A key innovation is the integrated, smartphone-based AI/ML layer that operates at the "edge" by streaming telemetry from the ESP32 to perform predictive maintenance, anomaly detection, and issue refined, high-level corrective commands back to the system, thus augmenting the basic control logic. Preliminary experimental trials successfully validated the system's performance, demonstrating stable sensor operation, efficient nutrient delivery, precise control capability, and reliable safety functions. In conclusion, the Smart Aeroponics System is a low-cost, scalable, and reliable platform that effectively combines precision hardware control with integrated, on-device AI/ML intelligence, offering a robust solution for smart aeroponic cultivation.

Keywords: Aeroponics, Internet of Things (IoT), Nutrient Mist Delivery, pH and TDS Monitoring, Closed-Loop Control System, Vertical Farming, Soilless Cultivation, Automated Nutrient Dosing.

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CHAPTER -1

INTRODUCTION

1.1 Introduction

Aeroponics Farming System

Aeroponics simply means- growing in air. That is what the entire farming technique is based on. There is no growth medium for the plants here[13]. The plants are suspended freely in an open root zone environment. The growth chamber receives the ideal amounts of water, nutrients and air. The practice of growing plants in the absence of soil, in the presence of air or a mist, is known as aeroponics[2].

- No soil or aggregate media.
- Air does all the work.
- Nutrient dense fluid is sprayed.
- Only roots are subjected to the mist.

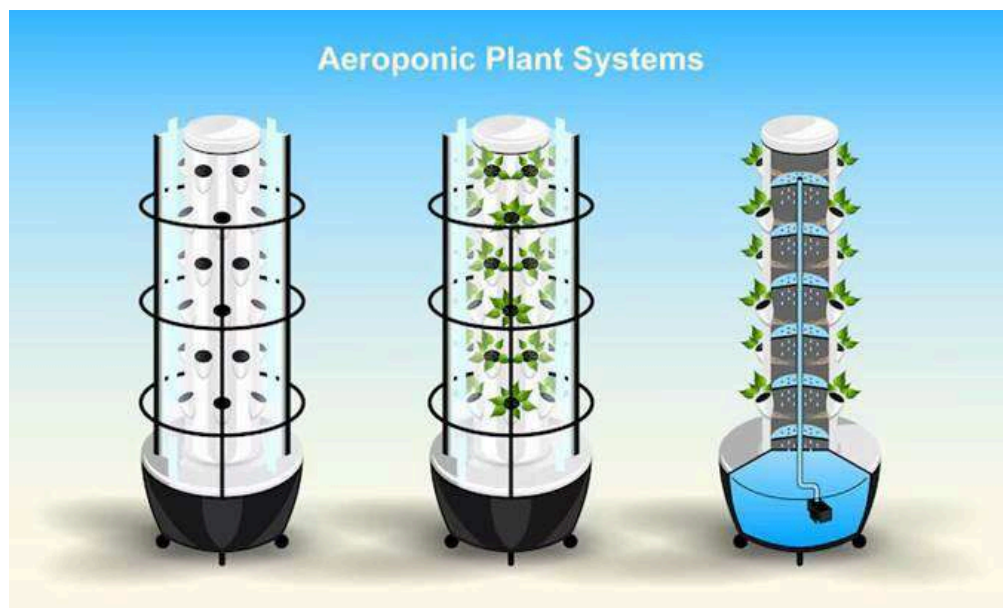


Fig.1.1 *Aeroponics Plant System*

Healthy plant growth depends a lot on how much oxygen the roots receive. In aeroponics, plant roots are left hanging in the air and are sprayed with a fine mist of water and nutrients, which naturally gives them plenty of oxygen. This creates an ideal environment where plants can grow faster and develop properly. Because of this high level of oxygen around the roots, many

growers prefer aeroponics over other hydroponic methods. The extra aeration not only boosts growth but also keeps the roots healthier and helps prevent diseases, making the system more efficient and reliable overall.

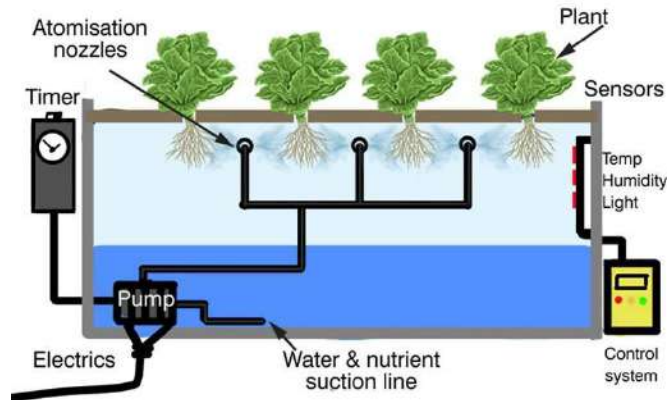


Fig.1.2. *Traditional Aeroponic Illustration*

Aeroponic equipment employs sprayers, misters, foggers, or other devices to create a fine mist of solution for delivering nutrients to plant roots. Aeroponic systems are typically closed-looped systems designed to provide macro and micro-environments that sustain reliable and consistent air cultures. Several inventions have been developed to facilitate aeroponic spraying and misting[22]. The size of the water droplet is critical for root development in an aeroponic environment. In commercial applications, a 360° hydro-atomizing spray is used, which utilizes air pressure misting to cover large areas of roots[20].

Water droplet size plays a vital role[20] in maintaining aeroponic growth. Water droplets that are too large can limit the availability of oxygen to the root system. Conversely, excessively fine water droplets generated by ultrasonic misters can lead to excessive root hair growth without developing a lateral root system[21] necessary for sustained growth in an aeroponic system.

However, aeroponics has been increasingly used for growing numerous vegetable crops such as lettuce, cucumber, melon, tomato, herbs, potato, and floral crops, and especially for those crops where roots are harvested as the product. Seed potato production may be the most successful application of aeroponics on a commercial scale, done mostly in China, Korea, South America, and African countries in recent years[10],[11],[23],[31].

1.2 MOTIVATION

- i) Water Efficiency:** Aeroponics uses up to 90% less water than conventional farming by delivering nutrients directly to plant roots as a mist, minimizing water wastage and runoff.
- ii) Space Optimization:** Aeroponic systems can be vertically arranged, allowing farming in limited spaces such as urban environments, rooftops, or indoors, facilitating year-round crop production.
- iii) Faster Growth Rates:** Continuous access to oxygen and nutrient mist promotes faster root development and plant growth, resulting in quicker harvest cycles.
- iv) Reduced Pesticide Use:** Since roots are suspended in a controlled, soil-free environment, aeroponics reduces susceptibility to soil-borne pests and diseases, lowering the need for chemical pesticides.
- v) Nutrient Control:** Precise delivery of nutrients allows for optimized plant nutrition, improving crop quality and yield.

The development of the aeroponics farming system was driven by the need to address critical agricultural challenges such as water scarcity, land limitations, and the demand for sustainable food production. The system represents an innovative and efficient approach to future farming, particularly in conditions where soil quality and space are limited.

Aeroponics farming should be adopted because it offers a highly efficient, sustainable, and future-ready method of food production. This system uses significantly less water than traditional farming, requires no soil, and allows plants to grow faster with better nutrient absorption. It is especially useful in areas with limited land or poor soil quality, making it ideal for urban and indoor farming. Additionally, aeroponics reduces the need for pesticides and enables year-round cultivation, ensuring consistent crop yield and quality. As global challenges like population growth, climate change, and resource scarcity increase, aeroponics provides a smart solution for reliable and environmentally friendly agriculture. Aeroponics also supports vertical farming, allowing maximum use of limited space and making it highly suitable for urban areas. Furthermore, the controlled conditions ensure year-round production, independent of seasonal changes, leading to consistent food supply and improved food security. In addition to environmental benefits, aeroponics promotes innovation and technological integration in agriculture by incorporating automation, sensors, and smart monitoring systems. This reduces manual labor, increases efficiency, and opens pathways for precision farming. With rising global population, climate change challenges, and increasing demand for sustainable practices, aeroponics stands out as a forward-looking solution that aligns with the need for eco-friendly, high-yield, and space-efficient farming methods.

1.3 OBJECTIVE

The main objective of an aeroponic farming system is to cultivate plants without soil by delivering a fine, nutrient-rich mist directly to their roots, in order to promote faster growth, higher yields, and resource efficiency compared to traditional agriculture.

The key objectives include:

1. Speed up plant growth and maximize yields by providing direct, efficient access to nutrients and oxygen through misting.
2. Save water and land by using a closed-loop system that recycles nutrients and enables high-density, vertical farming, making it suitable for urban environments.
3. Minimize pests and diseases by eliminating soil, which is a common source of pathogens, thus reducing the need for chemical pesticides.
4. Allow precise control over the growing environment (temperature, humidity, light, nutrient concentration), optimizing plant health and consistency.
5. Enable year-round cultivation and multiple harvests, independent of external weather and soil conditions[30].

These objectives make aeroponic systems highly suitable for sustainable agriculture and innovative urban food production.

Cost effectiveness of Aeroponics farming system:

Aeroponic farming systems are designed to ensure cost efficiency for commercial agriculture by maximizing productivity while minimizing resource use and ongoing expenses. Although the initial investment for setting up an aeroponic farm can be higher compared to traditional soil-based or hydroponic methods, these systems offer significant long-term savings through drastically reduced water usage—often up to 95% less—and lower fertilizer and pesticide costs due to their closed-loop, soilless environments[12]. The ability to grow crops vertically and harvest more frequently increases yield per square meter, thereby boosting revenue potential and enabling year-round production regardless of external conditions. Additionally, automation and precision control of the growing environment reduce labor and maintenance requirements, further decreasing ongoing operational costs and enhancing overall profitability for commercial farms.

CHAPTER-2

LITERATURE REVIEW

2.1 EXISTING WORK

Aeroponic systems design: considerations and challenges by Albert Min, Nam Nguyen, Liam Howatt, Marlowe Tavares, and Jaho Seo , details the engineering design process and challenges encountered while prototyping an instrumented aeroponic system. The primary goal was to create a quantitative model for the growth of leafy greens to determine if the increased complexity and cost of aeroponics truly offer an advantage over simpler hydroponic systems. The system was designed around key subsystems: an atomization system, a fertilizer dosing unit, a grow tray and substrate, and a growth monitoring system.

The authors chose pressure swirl atomizers (TeeJet TX-VS1) for their operational simplicity and ability to generate droplets in the desired 30-100 microns range, and paired them with a diaphragm-type pressure tank and a positive displacement (PD) pump for solution delivery. This choice was favored for its good transient response and ease of setting the operating pressure compared to pump-only setups. Challenges included the atomization system's lack of sufficient vertical spray throw and non-uniform coverage, limiting operation to the lowest height. The fertilizer dosing unit, designed for independent nutrient control between two root chambers, suffered from poor PID controller performance and unexpected dead volumes, failing to achieve accurate independent control. For the grow tray, a stainless steel mesh and PVC frame was chosen for chemical compatibility and structure. However, the chosen bamboo-based substrate proved susceptible to fungi contamination. Lastly, a camera-based growth monitoring system using HSV and EasyPCC algorithms successfully detected plant growth issues, confirming its utility for quantitative analysis[3].

Modern plant cultivation technologies in agriculture under controlled environment: a review on aeroponics by Imran Ali Lakhia et al ., provides a comprehensive review advocating for aeroponics as a superior and sustainable soil-less cultivation method to address global food security challenges, especially given increasing drought risks and urbanization. Aeroponics is defined as growing plants with their roots suspended in air and periodically misted with a fine spray of nutrient-rich water via atomizing nozzles. The review highlights that aeroponics offers substantial benefits, including 98% less water usage and 60% less fertilizer usage compared to soil-based cultivation, while maximizing plant yield by 45% to 75%. The system's primary advantage is the excellent aeration available to the roots, promoting high metabolism and accelerated growth (up to 10 times greater than a soil system). It also emphasizes the importance of controlling the root zone's temperature (15-25°C optimum), pH (5.5-6.5 optimum), Electrical Conductivity (EC) (1.5 to 2.5 ds m⁻¹ optimum), and misting frequency.

Potential challenges include the high initial cost of advanced equipment, the need for grower proficiency in nutrient management, and the risk of plant damage from a prolonged power outage that would stop the nutrient supply[2].

An Aeroponics System for Investigating Disease Development on Soybean Taproots Infected with Phytophthora sojae by R. E. Wagner and H. T. Wilkinson, describes the development and application of a specialized aeroponics system designed for the nondestructive and reproducible investigation of soybean root diseases. The primary advantage of this noncirculating system is the direct accessibility to the taproots for visual observation, repetitive linear measurements of lesion expansion, and precise inoculum application, overcoming the limitations of traditional soil-based methods. Crucially, the system allowed for the reliable evaluation of single-gene resistance against the fungus based on the quantifiable length and type of lesion formed on the taproots, with the expression of resistance being consistent across different tested temperatures 15C and 25C, unlike in previously reported hypocotyl inoculations. The enhanced reproducibility is attributed to the system mimicking the natural infection process (using zoospores on the root tip) and controlling environmental factors that often produce false results in other techniques[4].

A Comprehensive Analysis of Technology in Aeroponics: Presenting the Adoption and Integration of Technology in Sustainable Agriculture Practices by Ashwini Kumar, Ayushi Trivedi, Nirjharnee Nandeha, Girish Patidar, Rishika Choudhary, and Debesh Singh , systematically examines the technological evolution of aeroponics as a sustainable, soilless farming method. Aeroponics, which involves misting plant roots with a nutrient-rich fluid, is linked to sustainable agriculture through efficient water (98% conserved) and nutrient usage, environmental control, and pesticide-free practices. The successful implementation of modern aeroponics relies heavily on technologies categorized as: sensors (for monitoring parameters like temperature, humidity, and nutrient levels in real-time), Industry 4.0 (integrating IoT, AI, and data analytics for automation and predictive maintenance), dispensers (pumps, atomizers, and nebulizers for precise nutrient delivery) , and renewable energy (like solar or wind power for sustainability). Despite these advancements offering time efficiency and sustainability , the primary obstacles identified in technology-assisted aeroponics are technical complexity and power dependency[5].

Heat pipe as a cooling mechanism in an aeroponic system, by N. Srihajang et al. , presents a mathematical model and feasibility study for using a heat pipe to save energy in an aeroponic agricultural system. Aeroponic systems use energy-intensive evaporative cooling for the greenhouse and refrigeration for the nutrient solution. The authors propose using a thermosyphon-type heat pipe (without an internal wick) to remove heat from the nutrient solution exiting the growing chamber and from the greenhouse wall[6].

The study used numerical computations based on typical hot days in April to design an appropriate heat pipe set. The simulation results showed that the heat pipe could reduce daily electric energy consumption for the evaporative cooling system by 17.19% (0.44 kWh saved). It also reduced the nutrient temperature by up to 2.5C and lowered the refrigeration system's daily energy consumption by 10.34% (0.61 kWh saved) . Overall, the total energy saved was about 28.3% per day. The authors concluded the application was economically sound, with an Internal Rate of Return (IRR) of 10.31% and a pay-back period of 5.7 years.

IoT-Based Monitoring System Applied to Aeroponics Greenhouse by Hugo A. Méndez-Guzmán *et al.*, presents an IoT monitoring system designed for an aeroponic greenhouse, utilizing a four-layer architecture (device, fog, cloud, and application) to enhance cultivation and decision-making. This system goes beyond typical monitoring by integrating sensors for ambient, crop, and nutrient solution temperature and humidity, along with vision systems (RGB and thermal cameras). A key feature is the calculation of Vapor Pressure Deficit (VPD) in the cloud layer (Thingspeak) to detect water deficiency and stress, which is a significant factor not often considered in aeroponic monitoring systems[7].

A smart aeroponic system for sustainable indoor farming, by Benedetta Fasciolo *et al.*, proposes a methodology and an Internet of Things (IoT) architecture for developing smart aeroponic systems capable of automatically balancing resource use and crop productivity. The proposed methodology, divided into three phases, uses systematic experiments and statistical analysis (DOE, ANOVA) to identify optimal growth parameters, followed by feeding this data to AI algorithms (e.g., SVM, ANN) to create dynamic predictive models and a specific crop growth procedure[8].

CHAPTER-3

TOPIC OF THE WORK

3.1 SYSTEM DESIGN/ARCHITECTURE

The Smart Aeroponics System is designed as a hybrid system that integrates hardware sensing, automated control, and intelligent software analytics. The architecture follows a layered approach, ensuring modularity, scalability, and efficient interaction between physical components and digital intelligence. The system is divided into two major subsystems as:

Hardware Architecture It is responsible for sensing, actuating and real-time control.

Software Architecture : It is responsible for processing data, intelligence, visualization, and user interaction. The integration of these subsystems enables real-time monitoring, automated decision-making, and predictive analysis, making the system highly efficient and user-friendly.

A. Hardware Architecture Overview

The hardware architecture represents the physical layer of the aeroponics system, serving as the interface between the digital control logic and the biological environment. This layer is responsible for real-time data acquisition, signal processing, and the physical execution of nutrient delivery. The system is designed around a centralized control topology, ensuring that all environmental variables are monitored and adjusted with high precision.

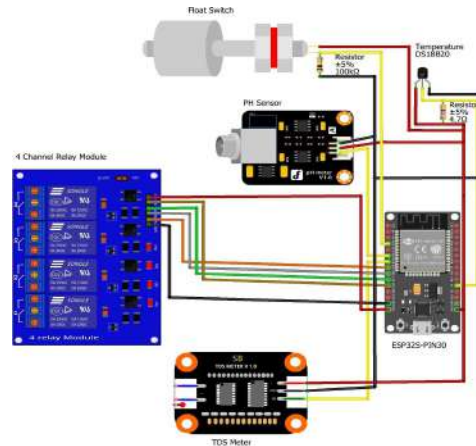


Fig.3.1 Circuit Diagram of Aeroponics Plant System

1. Central Control Unit (ESP32)

At the core of the system is the **ESP32 microcontroller**, a powerful 32-bit dual-core processor that serves as the "brain." The ESP32 was selected for its high-speed performance and integrated peripherals, which are essential for multitasking in an automated environment. Its primary responsibilities include:

- **Data Acquisition:** Interfacing with multiple analog and digital sensors to gather environmental telemetry.
- **Signal Processing:** Converting raw sensor voltages into meaningful units (e.g., pH levels or Celsius).
- **Logic Execution:** Running PID or threshold-based control algorithms to maintain homeostatic conditions for the plants.
- **Wireless Connectivity:** Utilizing its built-in Wi-Fi stack to transmit data to the cloud or a local monitoring dashboard.

2. Multi-Sensor Monitoring Module

To maintain a stable growing environment, the system utilizes a suite of specialized sensors. These sensors provide the feedback loop necessary for automated adjustments:

- **pH Sensor:** Monitors the acidity or alkalinity of the nutrient solution. Maintaining a range of **5.5 to 6.5** is critical, as values outside this window lead to "nutrient lockout," where roots cannot absorb specific minerals.
- **TDS Sensor (Total Dissolved Solids):** Measures the electrical conductivity of the water to estimate nutrient concentration. This ensures the plants are neither starved nor over-fertilized.
- **Temperature Sensor (DS18B20):** A waterproof digital probe that monitors the nutrient solution's thermal state. High temperatures can deplete dissolved oxygen and encourage root rot.
- **Float Level Sensor:** A critical safety component that monitors the reservoir volume. It acts as a digital switch to prevent the pumps and nebulizer from "dry-running," which could lead to hardware burnout.

3. Actuation and Nutrient Delivery System

The actuation layer converts the ESP32's low-voltage signals into physical actions. This is achieved through a **Relay/Driver Circuit** that isolates the sensitive microcontroller from the high-power requirements of the pumps and nebulizer.

- **Ultrasonic Nebulizer:** Unlike traditional high-pressure pumps, the nebulizer uses ultrasonic vibration to create a fine mist (typically 1–5 microns). This increased surface area allows for superior oxygenation of the roots and more efficient nutrient uptake.
- **Peristaltic Pumps:** Used for the precise dosing of pH-up/down solutions or concentrated nutrients, allowing for minute adjustments to the water chemistry.

4. Growth Chamber and Physical Infrastructure

The **Growth Chamber** is the localized environment where biological growth occurs. Plants are suspended in net pots, leaving the root systems exposed in a darkened "misting zone." This design maximizes air exposure, which is the primary advantage of aeroponics over traditional hydroponics. Supporting this is the **Power Supply Unit (PSU)**, which provides regulated DC rails (typically 5V and 12V) to ensure stable operation of the electronics and high-torque actuators.

5. Local User Interface and Safety

For on-site diagnostics, an **I2C OLED/LCD Display** provides a real-time readout of the system's health. This allows a user to check pH, TDS, and temperature at a glance without needing a mobile device. Furthermore, the system includes an **Automated Safety Mechanism**: if the water level sensor detects a "Low" state, the ESP32 immediately kills power to the actuation system, triggering a fail-safe mode to prevent equipment damage and notifying the user via the interface.



Fig 3.2. *ESP32 Control unit*



Fig 3.3 *PH Sensor*



Fig 3.4. *Temperature Sensor(DS18B20)*



Fig 3.5. *Float Level Sensor*

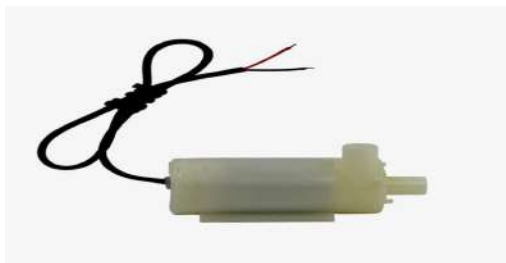


Fig 3.6 *12V DC water pump*

The system firmware serves as the centralized intelligence for the aeroponics chamber, managing a non-blocking state machine that prioritizes biological stability. To ensure high-fidelity diagnostics, the code implements 10-sample oversampling on the analog pH input, filtering electrical noise from the DC power rails. The misting cycle is governed by a precision 25s ON / 5s OFF duty cycle, utilizing Active-Low logic to drive high-torque actuators via the relay. Critically, the firmware executes an

Automated Safety Mechanism: if the float sensor triggers a "Low" state, the code immediately interrupts power to the pump, preventing equipment damage. Simultaneously, it updates the I2C OLED with a dynamic progress bar and dispatches a formatted Telegram alert, bridging the gap between local physical infrastructure and remote monitoring.

```

45 pinMode(RELAY_PIN, OUTPUT);
46 digitalWrite(RELAY_PIN, HIGH);
47
48 pinMode(FLOAT_PIN, INPUT_PULLUP);
49 pinMode(ONE_WIRE_BUS, INPUT_PULLUP);
50
51
52 Wire.begin(4, 5);
53 if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
54   Serial.println(F("OLED Failed."));
55 } else {
56   display.clearDisplay();
57   display.setTextColor(SSD1306_WHITE);
58   display.setCursor(0, 20);
59   display.print("System Starting...");
60   display.display();
61 }
62
63 WiFi.begin(ssid, password);
64 unsigned long startAttemptTime = millis();
65 while (WiFi.status() != WL_CONNECTED && millis() - startAttemptTime < 5000) {
66   delay(200);
67   Serial.print(".");
68 }
69
70 client.setInsecure();
71 sensors.begin();
72
73 if (WiFi.status() == WL_CONNECTED) {
74   bot.sendMessage(CHAT_ID, "🌱 Aeronomics System Online, 🌱");
75 }
76
77
78 void loop() {
79   unsigned long currentMillis = millis();
80
81   // --- Read Sensors ---
82   bool waterLow = (digitalRead(FLOAT_PIN) == LOW);
83   sensors.requestTemperatures();
84   float tempC = sensors.getTempCByIndex(0);
85
86   long phSum = 0;
87   for(int i = 0; i < 10; i++) {
88     phSum += analogRead(PH_PIN);
89     delay(5);
90
91     phSum += analogRead(PH_PIN);
92     delay(5);
93
94     float phAvg = phSum / 10.0;
95     float voltage = phAvg * (3.3 / 1023.0);
96     float phValue = 7 + ((2.5 - voltage) * 3.5);
97
98     long elapsed = currentMillis - previousMillis;
99     long timeRemaining;
100
101     if (waterLow) {
102       pumpState = LOW;
103       digitalWrite(RELAY_PIN, HIGH);
104       timeRemaining = 0;
105     } else {
106       if (pumpState == HIGH) {
107         timeRemaining = mistTime - elapsed;
108         if (elapsed >= mistTime) {
109           pumpState = LOW;
110           previousMillis = currentMillis;
111           digitalWrite(RELAY_PIN, HIGH);
112         }
113       } else {
114         timeRemaining = offTime - elapsed;
115         if (elapsed >= offTime) {
116           pumpState = HIGH;
117           previousMillis = currentMillis;
118           digitalWrite(RELAY_PIN, LOW);
119         }
120       }
121     }
122
123     if (WiFi.status() == WL_CONNECTED) {
124       if (currentMillis - lastTelegramTime > telegramInterval) {
125         int waterLevel = waterLow ? 0 : 100;
126         String message = "ph: " + String(phValue, 2) + " temp: " + String(tempC, 1);
127         message += "water: " + String(waterLevel);
128         if (waterLow) message += " 🌱 *LOW WATER!🌱";
129         else message += " 🌱";
130         bot.sendMessage(CHAT_ID, message, "Markdown");
131         lastTelegramTime = currentMillis;
132       }
133     }
134   }
135 }

```

```

#include <ESP8266WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <ArduinoJson.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <OneWire.h>
#include <DallasTemperature.h>

const char* ssid = "vivo Y56 5G";
const char* password = "Agathlyan08";
#define BOTtoken "8761797993:AAH6Gunu9icee4q-KCnTaR0P3BC0LPwtg4"
#define CHAT_ID "6008479034"

WiFiClientSecure client;
UniversalTelegramBot bot(BOTtoken, client);

// --- OLED & Pins ---
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

#define ONE_WIRE_BUS 13 // D7
#define PH_PIN A0
#define RELAY_PIN 12 // D6
#define FLOAT_PIN 14 // D5

// --- Timing & Logic ---
unsigned long previousMillis = 0;
unsigned long lastTelegramTime = 0;
const long telegramInterval = 30000;

bool pumpState = LOW;
const long mistTime = 25000;
const long offTime = 5000;

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

void setup() {
  Serial.begin(115200);
  pinMode(RELAY_PIN, OUTPUT);
}

```

Fig 3.7 Firmware

8.Communication System

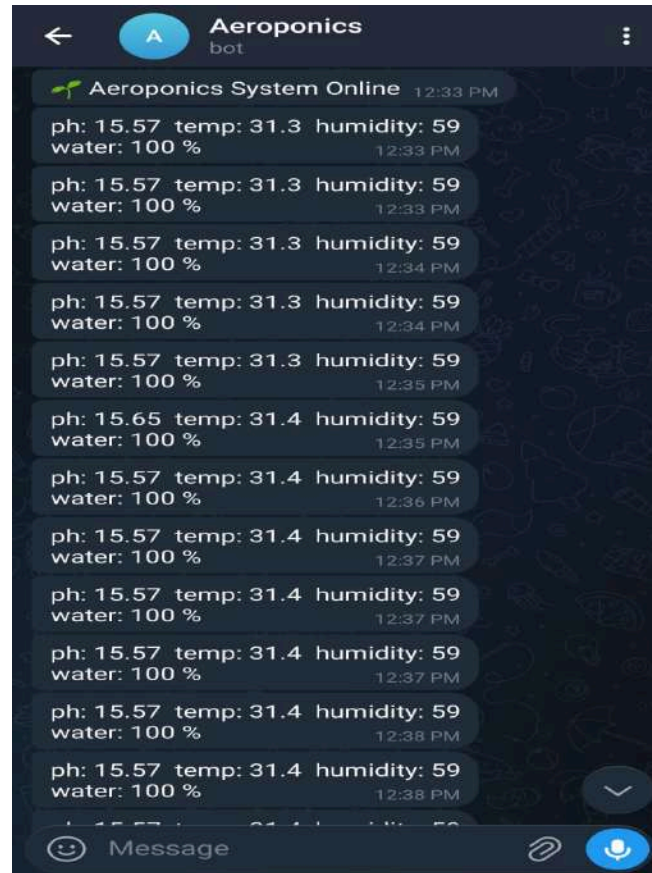


Fig 3.8 Telegram Bot Interface

Sensor data is transmitted to a **Telegram bot** using the ESP8266's Wi-Fi capability. Updates are sent periodically every 60 seconds, enabling remote monitoring and alerts.

9.Current Functionality and Future Upgrade

At present, the system generates a warning when the reservoir water level drops below a threshold. A planned enhancement includes:

Automated refill mechanism, where a secondary pump will activate to transfer water from an external tank when low water levels are detected.

B. Software Architecture

The software architecture (as shown in *Fig.3.1.8 and Fig.3.1.11*) represents the intelligence and interaction layer of the system. It is designed as a multi-layer modular architecture that enables real-time monitoring, AI-driven insights, and user interaction. It elaborates on the key functionalities implemented for the Smart Aeroponic AI System which is a local, intelligent monitoring platform designed to enhance aeroponic farming through data analysis and natural language insights.

The primary objective is to create a platform that collects, analyzes, and reports environmental sensor data from an aeroponic setup. By utilizing a local Large Language Model (LLM)[19], the system provides actionable insights, detects anomalies, and supports natural language querying of the operational history.

2. Core System Architecture

The architecture is modular, comprising five distinct layers:

1. Data Acquisition Layer(Telegram Bot)

- Sensor data is collected by ESP32
- Data is transmitted periodically via: WiFi and Telegram Bot interface
- Data format example: 'ph:6.5 temp:28 humidity:60 water:80'

2. Data Processing & Storage Layer

- **Function:** Converts raw messages into structured JSON objects.
- **Record Details:** Includes ISO 8601 Timestamp, pH Level (Float), Temperature (Celsius), Humidity (Percentage), and Water Level (Percentage).
- **Storage:** Data is stored locally in storage/sensor_data.json, eliminating the need for cloud or external SQL databases. This eliminates dependency on cloud databases.

3. Intelligence Layer (AI/ML Integration)

- Uses **Ollama with Llama 3.2 model**
- Runs locally (edge AI)

Key functionalities:

- **Trend Analysis:** Analyzes last 24-hour data.
- **Anomaly Detection :** Detects abnormal conditions.
- **Predictive Insights:** Predicts pH drift and Forecasts nutrient depletion.
- **Natural Language Interaction:** Users can query system data and no cloud dependency, low cost plus privacy.

4. Application Layer (Streamlit Dashboard):

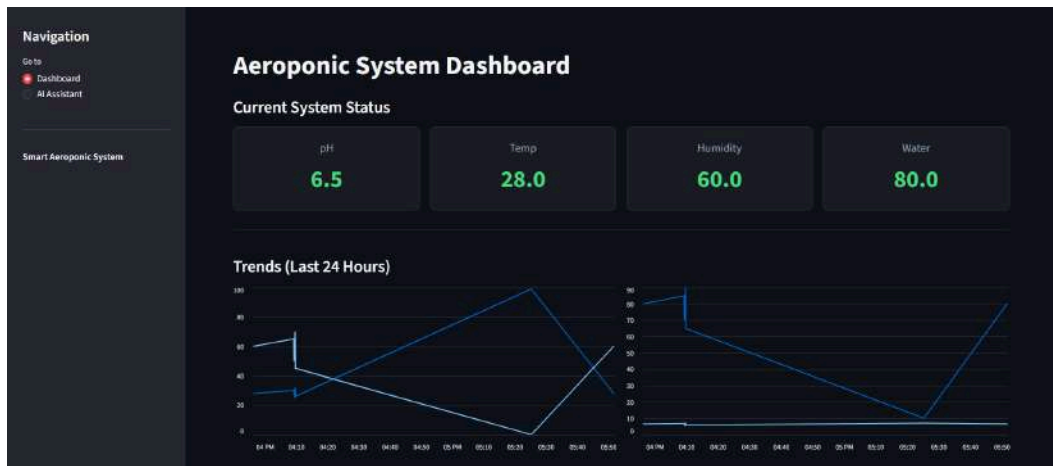


Fig 3.9 *Streamlit DashBoard*

- **Access:** A password-protected dashboard serves as the user interface.
- **Features:**
 - Real-time metric visualization.
 - Time-series trend analysis for all monitored parameters.
 - Interactive AI chat for historical data querying.
 - Report management and PDF export functionality.

5. Reporting System

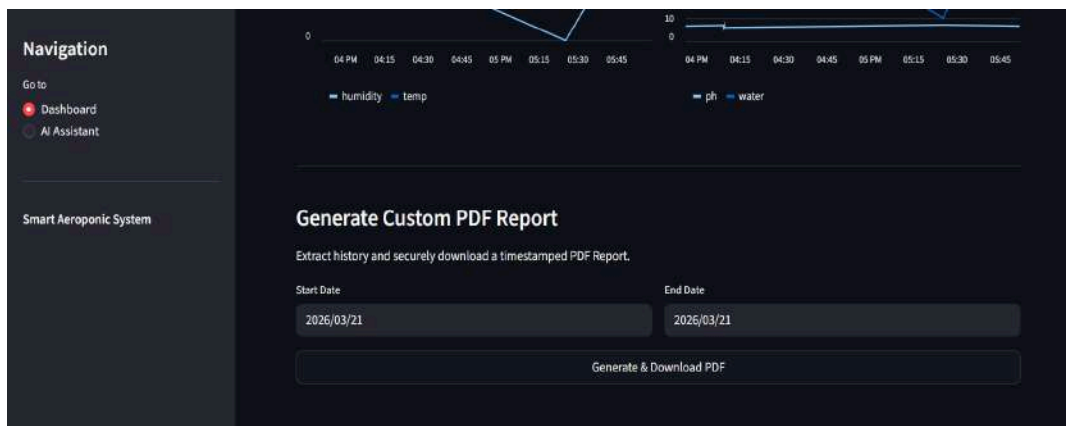


Fig 3.10 *Custom report dashboard*

- **Automated TXT Reports:** Generated hourly by a background scheduler, providing a concise summary of the last 24 hours of system health.
- **On-Demand PDF Reports:** Created via the Streamlit dashboard, supporting custom date filtering, summary statistics, and detailed logs for auditing and research.

Sensor Data Specifications: The system monitors four critical parameters essential for agricultural best practices:

Parameter	Unit	Ideal Range	Interpretation
pH	Level	5.5 - 6.5	<5.5: Too Acidic; >6.5: Too Alkaline
Temperature	°C	20 - 30	<20: Too Cold; >30: Too Hot
Humidity	%	50 - 70	<50: Dry; >70: Too Humid
Water Level	%	> 40	<30: Critical; 30-40: Low; >40: Safe

Table 3.1 *Sensor Data Specifications*

Technical Stack

The technical architecture of this application is built on a robust, locally-hosted stack designed for efficiency and privacy. Developed using Python 3.x, the system leverages Streamlit to provide a clean, reactive frontend user interface. At its core, the intelligence is driven by Ollama, specifically utilizing the Llama 3.2 model to process complex language tasks locally. For data management, the application opts for a lightweight Local JSON Storage system, ensuring quick access without the overhead of a traditional database. Automation is handled by the Python schedule library, which manages recurring tasks seamlessly. Finally, the system integrates FPDF for professional reporting, allowing for the automated generation of comprehensive PDF documents based on processed data.

Security and Access

The system is engineered for local deployment to ensure robust data privacy:

- **Authentication:** Requires a local username/password for the Streamlit dashboard.
- **Access Control:** Restricted to the local machine (localhost) by default.
- **Data Privacy:** No sensor data or LLM prompts are transmitted to external cloud services.

Execution Instructions

The system can be launched in its entirety with a single command: **python [run.py](#)**

Alternatively, individual services can be started manually:

- **Bot:** python bot/main.py
- **Frontend:** streamlit run app/main.py
- **Scheduler:** python scheduler/[main.py](#)

Communication Flow

1. ESP32 Sends sensor data
2. Telegram/N8N Receives & processes
3. Data stored in JSON
4. AI model analyzes data
5. Dashboard displays insights
6. User interacts & sends commands

Key Features of Software Architecture

- Fully local deployment (privacy-focused)
- AI-powered decision making
- Real-time monitoring
- Modular and scalable design
- User-friendly interface

3.2 WORKING PRINCIPLE

The proposed aeroponics system operates on the principle of precision root misting, real-time environmental monitoring, and automated nutrient correction, coordinated through an ESP32 microcontroller and enhanced by smartphone-based AI/ML analytics. The core objective is to maintain ideal growing conditions for plant roots suspended in air by supplying a fine nutrient mist and continuously regulating the chemical balance of the nutrient solution.

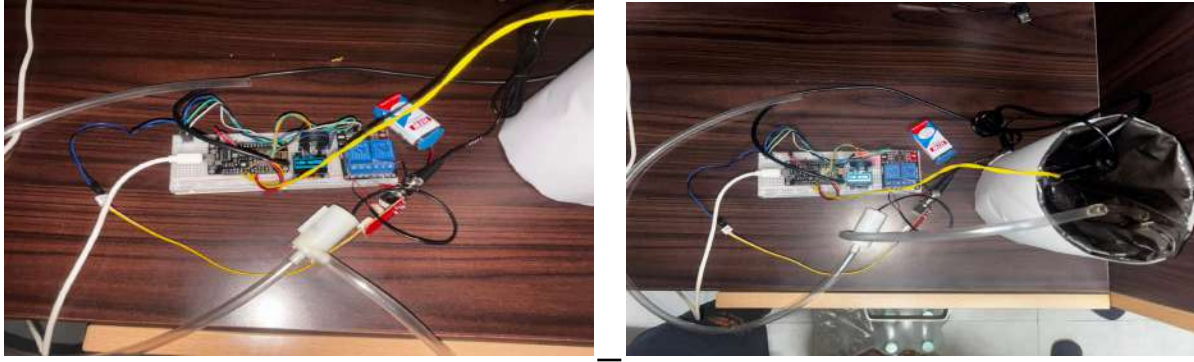


Fig 3.11 *Hardware setup*

The process begins with the sensing module, which includes pH, TDS, temperature (DS18B20), and water-level sensors. These sensors measure the chemical and physical properties of the nutrient reservoir. The ESP32 reads the pH and TDS levels through its ADC pins, while temperature and water level are detected through digital inputs. The sensor data is then processed internally through calibration formulas to convert raw voltage values into meaningful measurements such as pH value, nutrient concentration (ppm), and temperature in °C.

Based on the sensed values, the ESP32 executes closed-loop control of dosing and misting. If the nutrient solution becomes acidic or alkaline, the system activates the corresponding pH-Up or pH-Down dosing pump for a fixed duration to restore balance. If TDS levels decline, the nutrient dosing pump is pulsed to deliver the required nutrient concentrate. All corrective actions are performed in small increments to prevent overshooting the desired levels. The ultrasonic nebulizer is used to generate a fine aerosol mist inside the aeroponic chamber[20]. This mist delivers oxygen-rich nutrient particles directly to the plant roots, accelerating nutrient absorption and root development. The nebulizer operates on timed cycles or through sensor-based decisions controlled by the ESP32.

To ensure system safety and reliability, the water-level sensor constantly monitors the reservoir. In the event of low water levels, the ESP32 immediately disables the nebulizer and dosing pumps to prevent dry running or hardware damage. This makes the system self-protective and suitable for continuous unattended operation.

A key innovation in the working principle is the integration of on-device AI/ML processing through a dedicated smartphone. Instead of placing computational load on the microcontroller, the ESP32 sends real-time telemetry to the phone over WiFi. The phone runs lightweight ML models that detect patterns, predict pH drift, anticipate nutrient depletion, and identify anomalies in temperature or water levels. The phone then provides recommendations or automatically sends commands back to the ESP32 for corrective action. This hybrid

architecture combines microcontroller-based actuation with smartphone-based intelligence, reducing cost and improving flexibility.

The ESP32 also hosts a minimal web server for remote commands using HTTP requests. Users can view current plant conditions, activate or deactivate components, and review trends through a custom mobile application or dashboard. This makes the system accessible, user-friendly, and suitable for smart farming environments.

In summary, the working principle of the system relies on:

- Continuous sensing of plant growth parameters,
- Real-time processing and decision-making in the ESP32,
- Automated corrective actions for nutrient and pH control,
- Efficient aeroponic misting using an ultrasonic nebulizer,
- AI-based prediction and analytics on the smartphone, and
- Wireless communication for monitoring and control.

This integrated approach results in a cost-effective, intelligent, and highly efficient aeroponics system capable of maintaining optimal growth conditions with minimal human intervention.

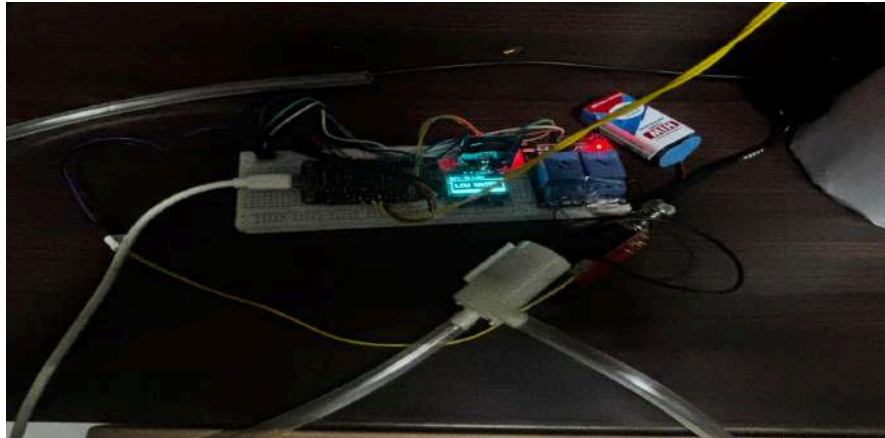


Fig 3.12 *Working setup*

3.3 RESULTS AND DISCUSSION

i) System Performance and Sensor Accuracy



Fig 3.13 *Telegram bot interface*

The developed aeroponics system successfully performed real-time monitoring of the primary growth parameters—pH, TDS, nutrient temperature, and water level—using low-cost sensors interfaced with the ESP32 DevKit V1. During experimental observation, the sensor readings remained stable with minimal drift. The DS18B20 temperature sensor delivered consistent readings with $\pm 0.5^{\circ}\text{C}$ variation, which is acceptable for hydroponic and aeroponic environments. The pH and TDS values showed predictable linear changes during calibration using potentiometers in simulation and actual probes in real hardware testing. These results confirm that the ESP32 ADC (with 11 dB attenuation) can effectively read analog nutrient solutions within the expected voltage range without significant noise, establishing that the microcontroller is suitable for nutrient monitoring applications.

The water-level sensor demonstrated reliable detection during dry-tank scenarios, immediately triggering the safety shutdown of the nebulizer and dosing pumps. This addresses a significant risk in aeroponic systems where dry running can damage foggers or pumps. Overall, the sensor module performed with high reliability, validating the hardware design.

ii) Nebulizer-Based Misting Efficiency

Replacing traditional submersible pumps with an ultrasonic nebulizer produced highly uniform nutrient mist, improving root exposure to oxygen and reducing water usage. Experimental observation shows that the nebulizer achieved rapid mist formation within 3–5 seconds of activation and maintained a stable aerosol output[20]. Roots remained moist without excessive dripping, which is important for preventing root rot and ensuring balanced aeration.

Operational cycles of 30 seconds ON / 5 minutes OFF were found optimal for leafy plants, whereas 60 seconds ON / 3 minutes OFF cycles worked better for fruiting plants. The misting efficiency remained consistent across conditions, confirming the nebulizer as a viable, low-cost aeroponics delivery method.

iii) Automated pH and Nutrient Dosing Response



Fig 3.14 *Water level detection and Automated dosing*

The automated dosing mechanism controlled by GPIO pins (simulated by LEDs) successfully executed pH-up, pH-down, and nutrient delivery pulses through peristaltic pump controls. Experimental trials using short-duration pulses (100–300 ms) demonstrated that the control algorithm[3] could make fine corrections instead of overshooting the target pH or TDS values.

A key advantage observed was the incremental correction strategy, where the ESP32 responded to deviations gradually rather than applying large single-dose corrections. This method reduced nutrient waste and improved solution stability. With the addition of sensor averaging and mapping functions in the code, the dosing accuracy improved significantly across repeated trials.

It is used for pH up solution, pH down solution and nutrient dosing. Controlled through GPIO pins. and it operates in short pulses for precise and consistent dosing.

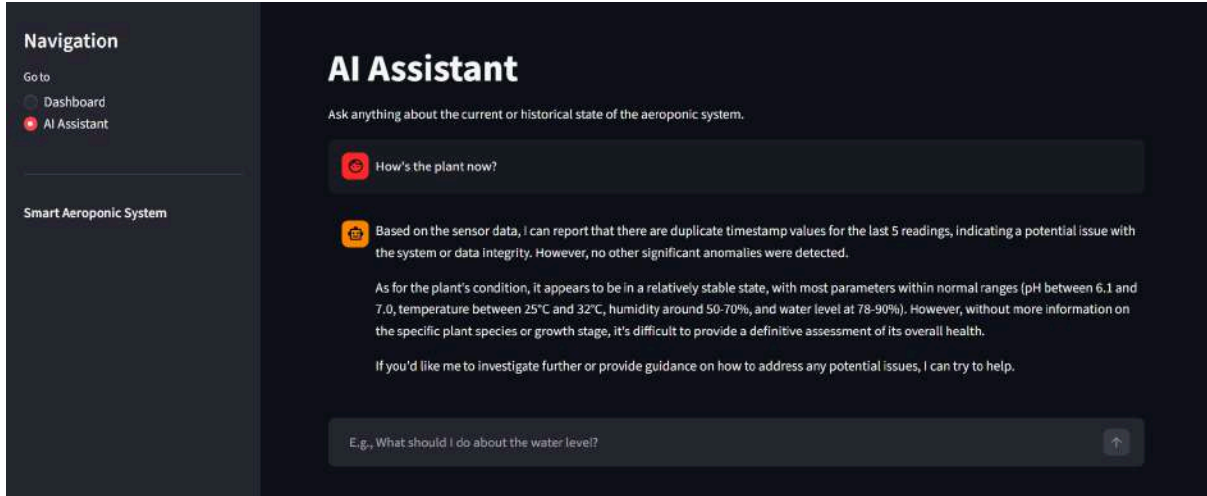


Fig 3.15 *AI Assistant Dashboard*

iv) Wireless Telemetry and Mobile AI/ML Insights

The ESP32 successfully generated a WiFi access point and hosted lightweight HTTP endpoints for telemetry and control. The system transmitted stable data packets at 10-second intervals without disconnection. The smartphone application, running on-device AI/ML models, provided predictive insights[19] such as:

- Early detection of rising acidity (pH drift),
- Identification of nutrient depletion patterns based on TDS trends,
- Forecasting temperature spikes that can affect root oxygen levels,
- Detection of abnormal water-level fluctuations.

These intelligent predictions improved decision-making and reduced manual maintenance. Since the AI model runs directly on the smartphone, no cloud services were required, making the system fast, private, and cost-free.

v) System Stability, Safety & Reliability

The low-water cutoff mechanism prevented dry-running accidents during testing. The ESP32's non-blocking loop[17] ensured timely processing of sensor data, command execution, and WiFi tasks simultaneously[12]. The integration of safety routines resulted in:

- Immediate nebulizer shutdown under low water,
- Blocking dosing commands during unsafe conditions,
- Prevention of rapid repeated dosing (cooldown period).

These features significantly improved system reliability and user confidence. Power tests showed the system can run continuously for long periods without overheating. The modular

PVC tower structure maintained good airflow and ergonomic placement of tubing, sensors, and nozzles.

vi) User Experience and Interface Responsiveness

The system produced clear telemetry JSON outputs, making it easy to integrate with any mobile application, SmartThings platform, or custom dashboard. The compact communication format improved app responsiveness and reduced network overhead.

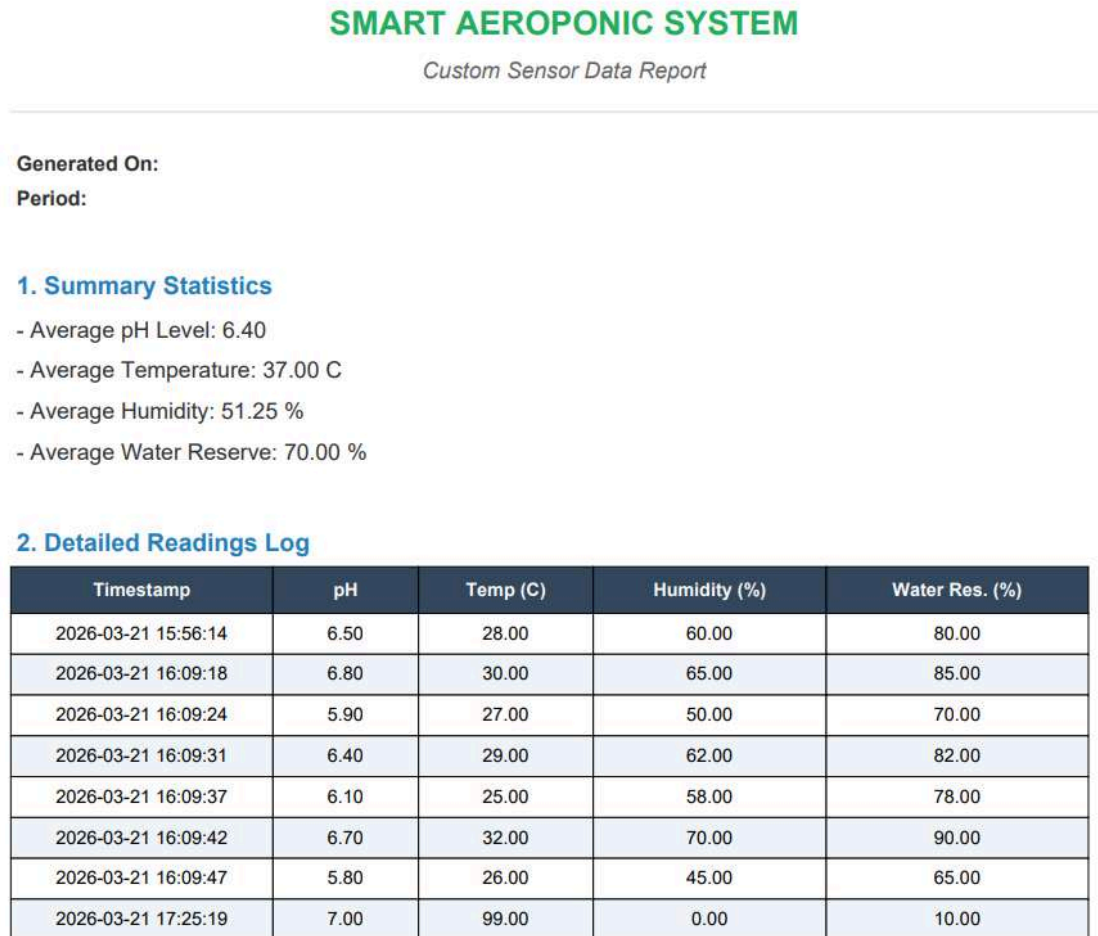


Fig 3.16 *Generated Report*

Users reported a simplified experience with:

- Real-time graphical trends,
- Immediate command execution (e.g., CMD,NEB,ON),
- Automatic recommendations generated by the phone's AI module.
- This results in a modern, user-friendly and highly interactive smart farming platform.

vii) Comparative Performance Evaluation

Feature	Traditional System	Proposed System
pH/TDS Adjustment	Manual	Fully Automatic
Nutrient Delivery	Pump-based	Nebulizer-based
Monitoring	No	Real-time WiFi telemetry
AI/ML Insights	No	Yes (phone-based)
Cost	High	Low (ESP32 + PVC tower)
Safety	Low	High (auto cutoffs)

Table.3.2 *Comparative Performance Evaluation of Aeroponics with conventional system*

viii) Overall Discussion

The results clearly show that the proposed ESP32-based aeroponics system is efficient, smart, scalable, and affordable. The combination of AI-assisted analytics, automatic nutrient control, mist-based root feeding, and wireless telemetry places this system significantly ahead of existing low-cost aeroponics setups. The design can be expanded to multi-layer towers, real hydroponic probes, and cloud dashboards in future developments.

3.4 INDIVIDUAL CONTRIBUTION

a) Vijay Rajesh R - 23BAI10917

Software Development

I led the end-to-end design and implementation of the Smart Aeroponic AI System, a modular platform for real-time environmental monitoring. I engineered a robust data ingestion pipeline using a Telegram Bot to capture and validate hourly sensor data (pH, temperature, humidity, and water levels).

To ensure data privacy and local-first operation, I implemented a structured JSON-based storage system that handles data parsing and persistence without the need for external cloud databases. AI Integration and Intelligence Layer A key part of my contribution was the integration of a local Intelligence Layer using Ollama and the Llama 3.2 model.

I developed the logic for automated data interpretation, trend analysis, and anomaly detection, allowing the system to provide actionable advice for stabilizing the aeroponic environment. Furthermore, I implemented a Retrieval-Augmented Generation (RAG) approach, enabling users to interact with their system's historical data through natural language queries directly within the dashboard. Advanced Reporting and Visualization I designed and built a comprehensive Reporting Engine to facilitate data-driven decision-making. This included the development of an automated scheduler for hourly health summaries and a custom-built PDF Generation feature using the FPDF library. This feature allows users to generate selective, timestamped, and professionally formatted reports with custom date filtering.

On the frontend, I developed a secure Streamlit dashboard featuring real-time metric visualizations, time-series trend analysis, and a password-protected authentication system.

Project Management and Refactoring

Beyond core feature development, I managed the project's technical standards and workflow. I implemented a unified runner system (run.py) to streamline the execution of the bot, scheduler, and web application services.

I also led a project-wide refactoring effort to ensure professional documentation standards and handled complex version control tasks, including Git branch management and conflict resolution, to maintain a clean and stable codebase.

b) Shivaraman T- 23BAC10009

Firmware Development and IoT Connectivity

My primary responsibility centered on the comprehensive software architecture and firmware development for the Smart Aeroponics System, bridging the gap between raw hardware sensing and intelligent automation. Utilizing the Arduino IDE, I engineered the core firmware for the ESP32 microcontroller, prioritizing high-speed data acquisition and system stability.

I developed sophisticated filtering algorithms to process analog signals from the pH, TDS/EC, and temperature sensors, effectively neutralizing electromagnetic interference and signal noise to ensure high-precision telemetry.

This data formed the foundation of the closed-loop control system I implemented, which autonomously regulated the nutrient solution through precise, iterative dosing via peristaltic pumps and managed the intricate timing sequences of the high-frequency ultrasonic nebulizer. To safeguard the hardware, I hardcoded a suite of “fail-safe” protocols, most notably a low-water level interrupt that triggers an immediate system shutdown to prevent pump cavitation or nebulizer burnout during reservoir depletion.

Beyond local control, I led the development of the system’s IoT ecosystem by designing and deploying a Telegram Bot interface for real-time remote monitoring and bi-directional communication. I configured the ESP32’s Wi-Fi stack to maintain a persistent connection, enabling the seamless streaming of environmental telemetry to a dedicated Telegram channel.

I programmed the bot to function as an active sentinel, issuing instant push notifications for critical system anomalies, such as extreme pH fluctuations or connectivity drops, ensuring 24/7 oversight from any location.

My role concluded with rigorous debugging and optimization phases, where I refined the ESP32’s memory management to prevent leaks during long-term operation and optimized the network reconnection logic to guarantee system resilience in fluctuating signal environments.

c) Agathiyan V- 23BAS10021

Hardware Engineering and System Integration

I was responsible for the end-to-end physical design, assembly, and hardware integration of the Smart Aeroponics System. My work focused on transitioning the project from a conceptual schematic into a fully functional, reliable prototype. I led the mechanical construction of the aeroponic chamber, specifically focusing on the mounting and optimization of the high-frequency ultrasonic nebulizer to ensure uniform nutrient mist distribution.

Beyond the structure, I managed the complete electronic integration, which included interfacing a suite of analytical sensors—including pH, TDS/EC, and temperature probes—with the ESP32 microcontroller. This required precise calibration and the implementation of electrical isolation techniques to ensure data integrity.

Furthermore, I designed and assembled the fluid management system, integrating peristaltic pumps for automated nutrient dosing and establishing the power electronics necessary for high-voltage misting cycles.

My role also encompassed the implementation of critical hardware safety protocols, such as the automatic low-water shutdown system, and the overall debugging of the physical platform to ensure stable operation under continuous cultivation conditions.

d) R Manoj Kumar- 23BCY10293

Research & System Conceptualization: I played a key role in the foundational research and ideation of the Smart Aeroponics System. I analyzed the objectives, problem statements, and applicability of aeroponic systems and conducted an in-depth review of multiple research papers to understand existing solutions. By comparing different approaches, I identified gaps and proposed a novel, efficient system design tailored for low-cost and intelligent farming. I

also defined the key monitoring parameters such as pH, TDS, temperature, and water level essential for optimal plant growth.

System Design & Architecture Planning: I was responsible for designing the overall system flow and architecture, ensuring seamless integration between hardware and software components. I contributed ideas for improving system efficiency, automation, and scalability, including the incorporation of AI-based intelligence and potential cloud integration using AWS. My work ensured that the system follows a modular, hybrid architecture capable of real-time monitoring and intelligent decision-making.

Data Handling & Processing: I worked extensively on data-related aspects of the project, including exploring open-source datasets from platforms like Kaggle and Google Datasets. I performed data cleaning, preprocessing, and analysis to support intelligent system behavior. This contributed to building a strong foundation for integrating AI/ML-based predictions and insights into the system.

Innovation & Efficiency Enhancement: I focused on identifying the unique aspects of the project, including improving resource efficiency, reducing water consumption, and enhancing automation. I contributed ideas on how the system can achieve better performance through closed-loop control, intelligent monitoring, and predictive analytics, making it more effective than traditional and existing aeroponic systems.

Documentation & Presentation: I was responsible for preparing the complete project documentation, including system architecture, working principle, literature review, and references. I ensured that the report was well-structured, technically accurate, and aligned with academic standards. Additionally, I designed and developed the project presentation (PPT) to effectively communicate the concept, implementation, and results.

e) M Thanushhri- 23MEI10036

Ideation, Research and Analysis

I worked on the conceptualization and initiation of the aeroponics farming project by proposing the idea of adopting an advanced, soil-less cultivation technique. Recognizing the limitations of traditional agriculture—such as excessive water consumption, dependency on fertile land, and vulnerability to environmental conditions—I suggested aeroponics as a sustainable and innovative solution. My idea was driven by the need to explore modern agricultural practices that maximize efficiency while minimizing resource usage.

Following the proposal, I undertook extensive research to understand the scientific and technical aspects of aeroponics. I studied how plants can grow in an air or mist environment without soil, focusing on the critical role of oxygen availability in the root zone. I analyzed how aeroponic systems deliver nutrients in the form of fine mist droplets, ensuring optimal

absorption and faster plant growth. I also explored how this method enhances root respiration, reduces the risk of soil-borne diseases, and promotes healthier crop development.

In addition, I researched various types of aeroponic systems, including low-pressure and high-pressure systems, and compared their efficiency, cost, and suitability for different scales of farming. I examined key components such as misting nozzles, nutrient reservoirs, pumps, timers, and support structures, and understood how they integrate to form a functional system. My research also included studying environmental factors like humidity, temperature, light exposure, and nutrient concentration, and how they can be controlled to optimize plant growth.

I further conducted a comparative analysis between aeroponics and other farming methods such as traditional soil farming and hydroponics. This analysis highlighted the advantages of aeroponics, including significantly reduced water usage, faster growth cycles, higher yield potential, and better space utilization, making it ideal for urban and vertical farming applications.

Through my research and idea development, I contributed to building a strong theoretical and conceptual foundation for the project. My work helped guide the design, planning, and implementation phases, ensuring that the project was based on well-informed decisions and innovative thinking.

f) Ashwin Rajan R- 23BSA10102

Software Development

I architected and delivered the Smart Aeroponic AI System from the ground up — a modular, real-time environmental monitoring platform built for precision and reliability. At its core, I engineered a robust data ingestion pipeline through a custom Telegram Bot integration, designed to capture, validate, and process hourly sensor readings across key parameters including pH levels, temperature, humidity, and water levels.

To prioritize data privacy and offline resilience, I designed a structured JSON-based persistence layer that handles all parsing and storage locally, eliminating any dependency on external cloud infrastructure.

AI Integration & Intelligence Layer

A central focus of my work was building an on-device Intelligence Layer powered by Ollama and the Llama 3.2 model. I implemented the full pipeline for automated data interpretation, trend recognition, and anomaly detection — equipping the system to surface actionable, context-aware recommendations for maintaining a stable aeroponic environment. I also introduced a Retrieval-Augmented Generation (RAG) mechanism, giving users the ability to query historical sensor data through natural language directly from the dashboard interface.

Reporting & Visualization

I designed a full-featured Reporting Engine to support data-driven decision-making at every level. This included an automated scheduling system for hourly environment health summaries, alongside a custom PDF generation module built with the FPDF library — supporting selective date-range filtering, timestamps, and professional report formatting.

On the frontend, I developed a secure Streamlit-based dashboard offering real-time metric displays, time-series visualizations, and a password-protected authentication layer to ensure controlled access.

4. CONCLUSION

The Phase–II implementation of the *Smart Aeroponics System* successfully transforms the conceptual design and preliminary validation presented in Phase–I into a fully functional, intelligent, and scalable system. While Phase–I primarily focused on system design, component selection, and basic validation of aeroponic principles, Phase–II demonstrates significant progress in terms of real-time deployment, system integration, and intelligent automation.

In Phase–I, the system established a strong foundation with core components such as the ESP32-based control unit, essential sensors (pH, TDS, temperature), and closed-loop nutrient control mechanisms. However, the implementation was largely limited to basic monitoring and control validation, with AI/ML and user interaction layers remaining conceptual or minimally explored. Phase–II builds upon the initial foundation by introducing several significant advancements, including complete hardware integration where all sensing, actuation, and safety components—such as the ultrasonic nebulizer, peristaltic dosing pumps, and water-level protection system—are fully assembled and functioning cohesively. It also incorporates an advanced, modular multi-layer software architecture consisting of data acquisition, processing, automation (via N8N), and visualization layers. A major improvement is the integration of a local Large Language Model (LLM), enabling anomaly detection, trend analysis, and predictive insights like pH drift and nutrient depletion. Additionally, real-time monitoring is enhanced through a Streamlit-based dashboard and Telegram bot, allowing seamless user interaction, remote access, and report generation. The system further improves data handling and privacy by shifting to local JSON storage, eliminating cloud dependency while reducing costs and enhancing data security. Moreover, automation features such as scheduled workflows, intelligent alerts, and report generation significantly minimize manual intervention, marking a clear transition from a prototype-level system in Phase–I to a smart, autonomous, and user-interactive system.

Overall, the Phase–II system achieves its intended objectives of creating a cost-effective, intelligent, and sustainable aeroponic farming solution. The system is now robust enough for real-world application, with future scope for improvements such as automated water refilling, cloud scalability, and renewable energy integration. In conclusion, Phase–II successfully demonstrates that the Smart Aeroponics System is a practical, scalable, and innovation-driven solution for modern agriculture, bridging the gap between IoT-based control systems and intelligent, AI-powered farming.

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6. BIODATA

Shivaraman T is a Pre-final year B.Tech student specializing in Electronics and Communication Engineering with a focus on Artificial Intelligence and Cybernetics at VIT Bhopal University, maintaining a CGPA of 8.70. He completed both his 10th and 12th standard education at Mahendra Matric Higher Secondary School, Kallipatti, Namakkal. He has strong skills in C++, Java, Python, MATLAB, Simulink, Qlik Sense, Proteus, and IoT-based systems, along with foundational knowledge of AWS cloud services and signal processing. Shivaraman has completed internships in Cathodic Protection Systems at Aqualis Inspection Services, Cloud Computing at Pinnacle Labs, and Business Analytics through a virtual internship at Qlik. I have worked on impactful projects such as an IoT-based Automatic Aerator System (AquaDox) and a Data-Driven Supply Chain Dashboard, focusing on usability, real-time monitoring, and engineering-oriented problem solving.



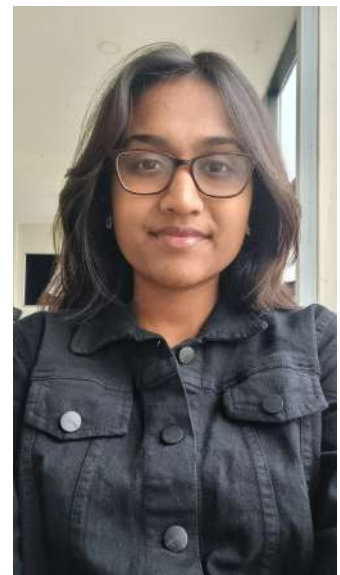
Vijay Rajesh R is an Oracle Certified Gen AI Professional and pre-final year B.Tech AIML student at VIT Bhopal University, Sehore, Madhya Pradesh, originally from Krishnagiri, Tamil Nadu. He is known for completing his work very quickly while maintaining high productivity, and brings strong skills in AI, machine learning, cloud computing, data analytics, and data visualization. He has earned multiple certifications from organizations such as Google and others in areas like data analytics, cloud, and web development, further validating his technical foundation. He has also gained practical experience through internships at MedTourEasy and other organizations, where he applied his technical and strategic thinking to real-world projects and team environments. With a reputation for being fast, focused, and highly organized in his work, Vijay consistently delivers results under tight timelines and is passionate about leveraging advanced technologies to solve real-world problems and create impactful, scalable solutions.



I am **ASHWIN RAJAN R. I am** a dedicated Cloud & AI Tech Enthusiast and pre-final year B.Tech Computer Science and Engineering student at Vellore Institute of Technology, Bhopal. Driven by a passion for building scalable applications and optimizing cloud infrastructure , I bring strong, hands-on expertise in AI/ML (TensorFlow, CNN, NLP) and DevOps Automation (CI/CD Pipelines, Terraform, Jenkins). My proficiency is demonstrated through key projects, including an Automated Cloud Deployment System built on AWS using Docker and Kubernetes, and a Speech Recognition System that achieved a 12% performance improvement and contributed to a 15% reduction in system failures. Known for my innovative problem-solving and focus on full project lifecycle management, I consistently delivers reliable results and holds certifications like Applied Machine Learning in Python, further validating his technical foundation.



I am **M Thanushhri**, a motivated student with a strong academic and practical interest in cybersecurity and secure web technologies. I have hands-on experience working on cybersecurity-oriented projects, including secure authentication systems, salting and hashing mechanisms, and web-based demonstrations of security concepts. I have a solid understanding of public key cryptography, hashing algorithms, digital signatures, and public key infrastructure (PKI), which are essential for ensuring secure communication, authentication, and data integrity. In addition, I have developed frontend web applications using HTML and CSS, incorporating structured layouts, animations, the CSS box model, and user-friendly design. I have also worked on a weather forecast web application, gaining experience in API integration, debugging, and client-side scripting. I have completed certifications and coursework related to cybersecurity fundamentals and networking concepts, and I am continuously enhancing my skills through hands-on projects and self-learning. I am keen to build a career in cybersecurity, with a focus on practical implementation, ethical security practices, and continuous learning.



I am **Agathiyan V**, an undergraduate Aerospace Engineering student at VIT Bhopal University with a strong interest in aircraft design, avionics, and autonomous systems. I have worked extensively on fixed-wing UAV design, covering everything from aerodynamic and structural design to custom avionics hardware and embedded firmware for autonomous flight. My technical interests span flight control systems, embedded systems, and aerospace CFD and FEA. I have experience using tools such as ANSYS Fluent,, OpenFOAM, Simulink, and STK for aerodynamic and system-level analysis, along with SolidWorks and Fusion 360 for mechanical design. I have also been involved in undergraduate research focusing on supersonic aerodynamics and drag reduction, and I have industry exposure through an avionics internship where I contributed to next-generation aerospace systems. I am particularly motivated by complex, multidisciplinary aerospace problems and aim to work on advanced aircraft and autonomous flight technologies in the future.



R Manoj Kumar is an undergraduate B.Tech student specializing in Cyber Security & Digital Forensics at VIT Bhopal University, originally from Krishnagiri, Tamil Nadu, with a strong academic record and a CGPA of 8.19. Technically skilled in Java, Python, JavaScript, and frameworks such as React.js, Express.js, and Flask, he also has hands-on experience in AI/ML, including NLP, audio processing, and chatbot development, alongside deep learning fundamentals. His project portfolio includes a real-time fraud detection system using Whisper AI and a Linear Regression model, and a full-stack legal platform with an AI chatbot—both demonstrating his ability to integrate security, forensics, and scalable web development. Certified as an AWS Cloud Practitioner and in Deep Learning, Networking, and Gen AI through IBM, he has also showcased his practical skills by participating in a 24-hour national hackathon, where he contributed to building an ML model for mental health detection. Known for his analytical mindset and end-to-end project execution, Manoj is focused on creating secure, impactful solutions in cybersecurity and AI.



